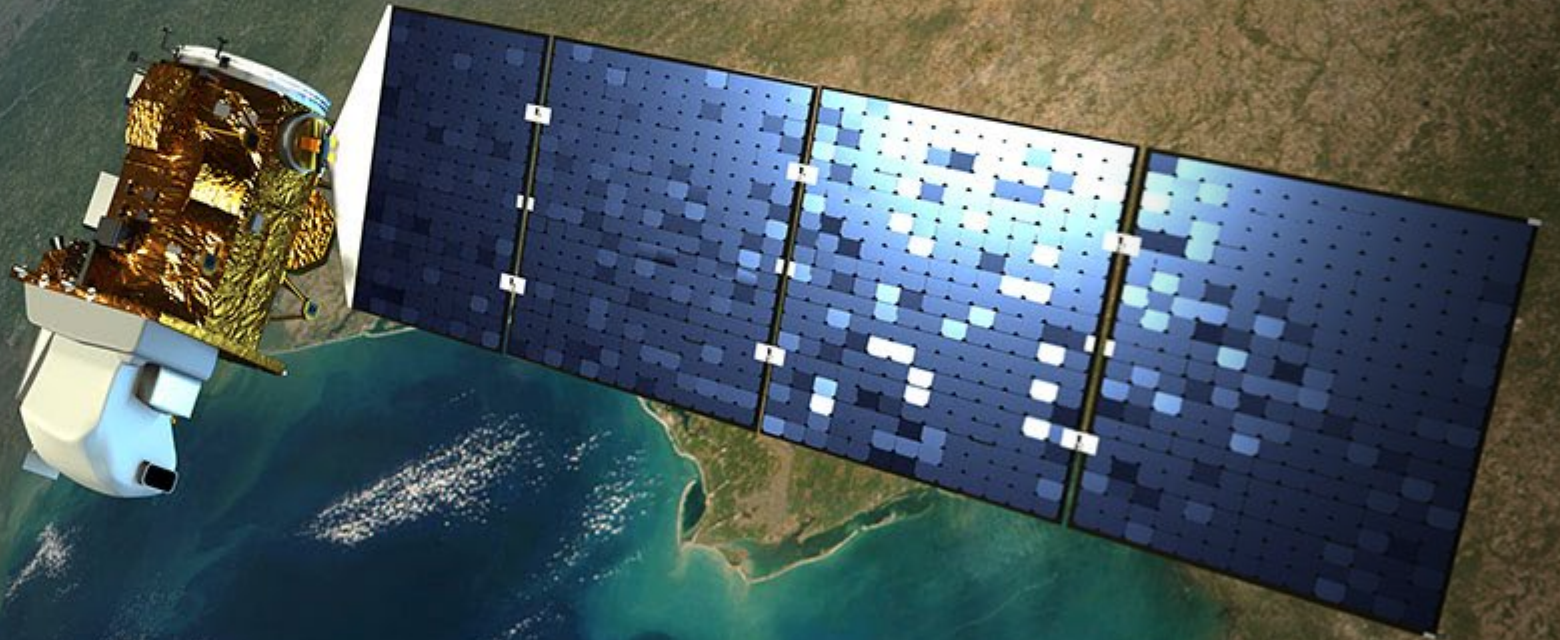
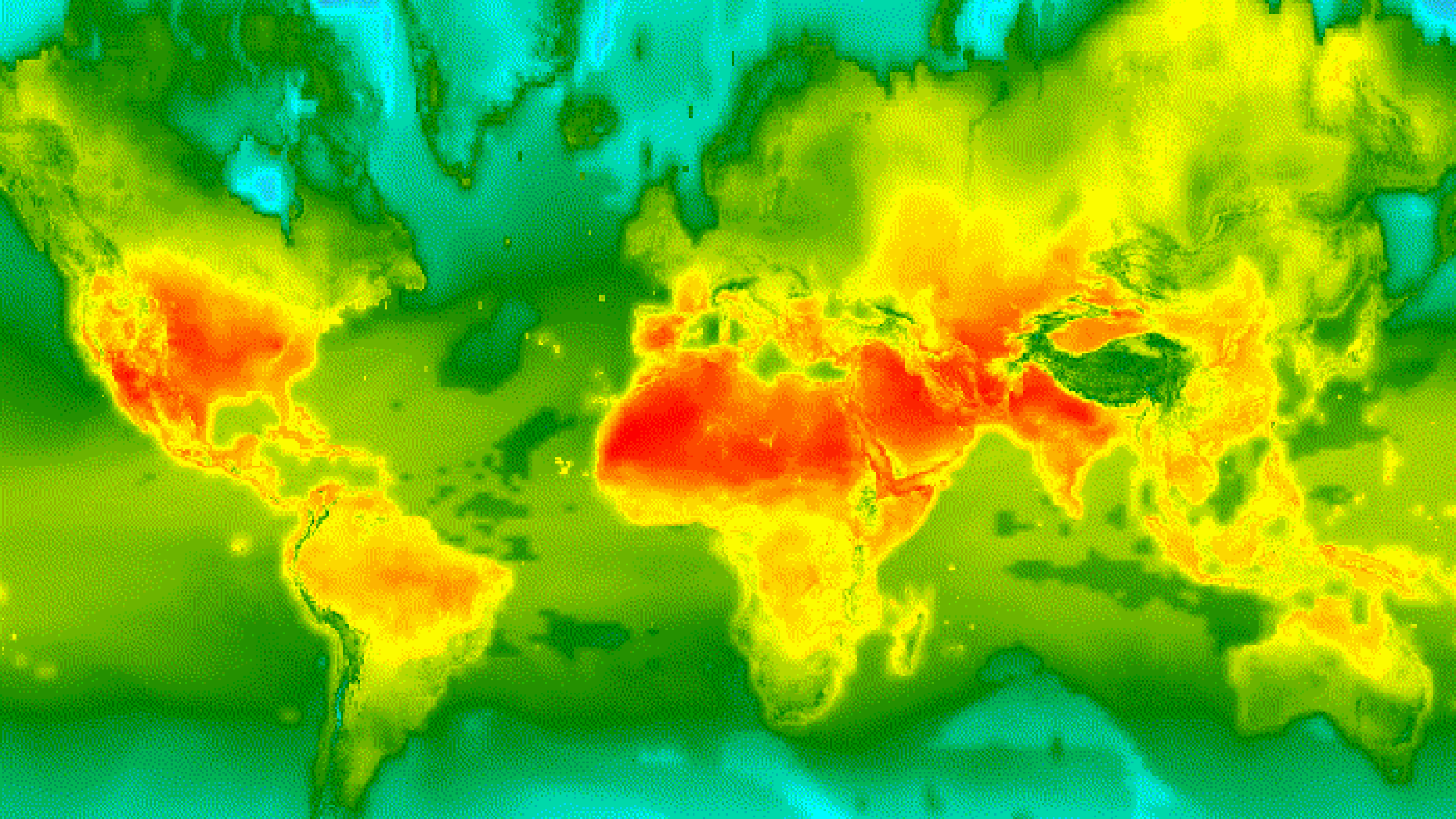
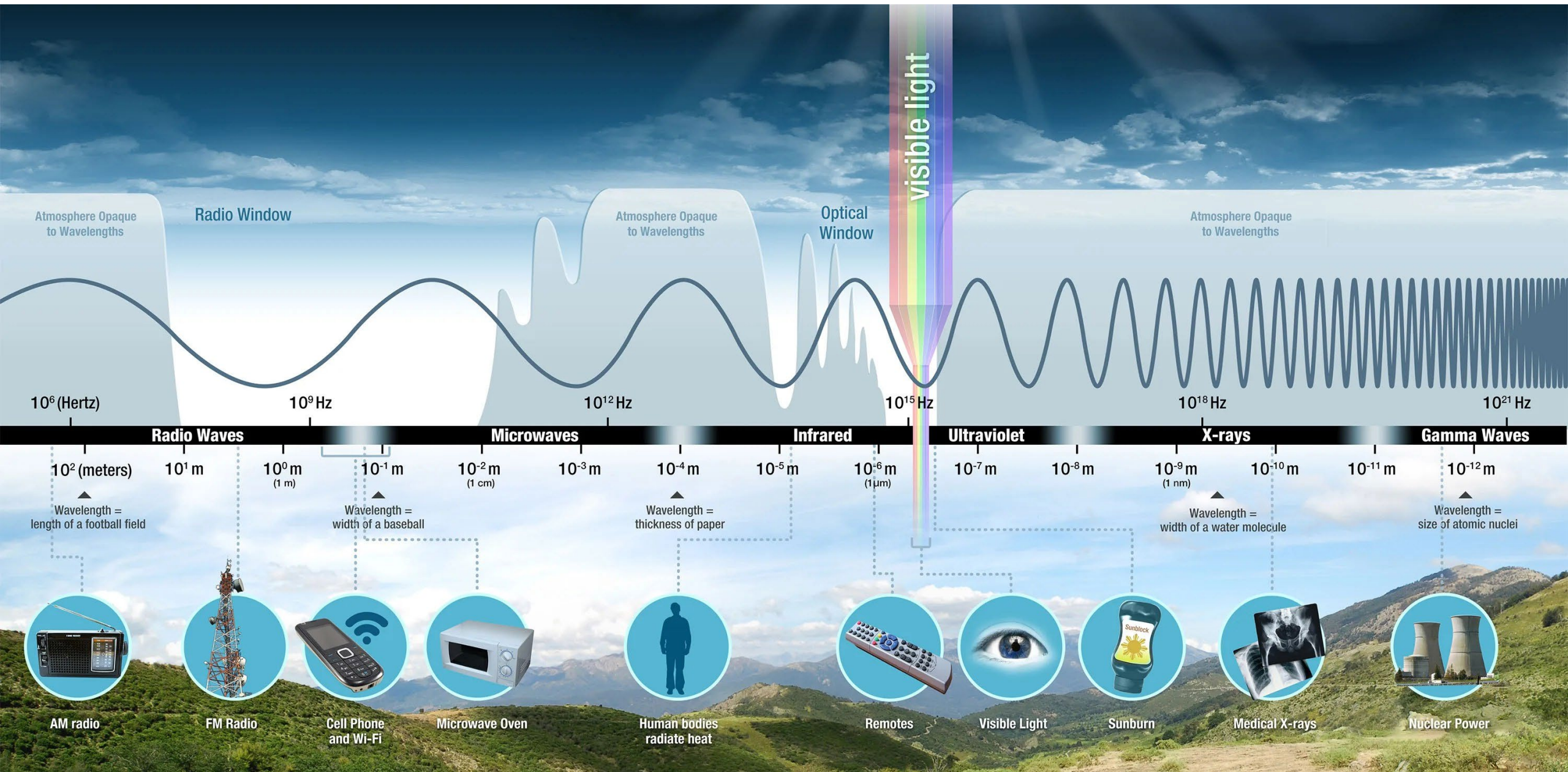


Remote Sensing Applications

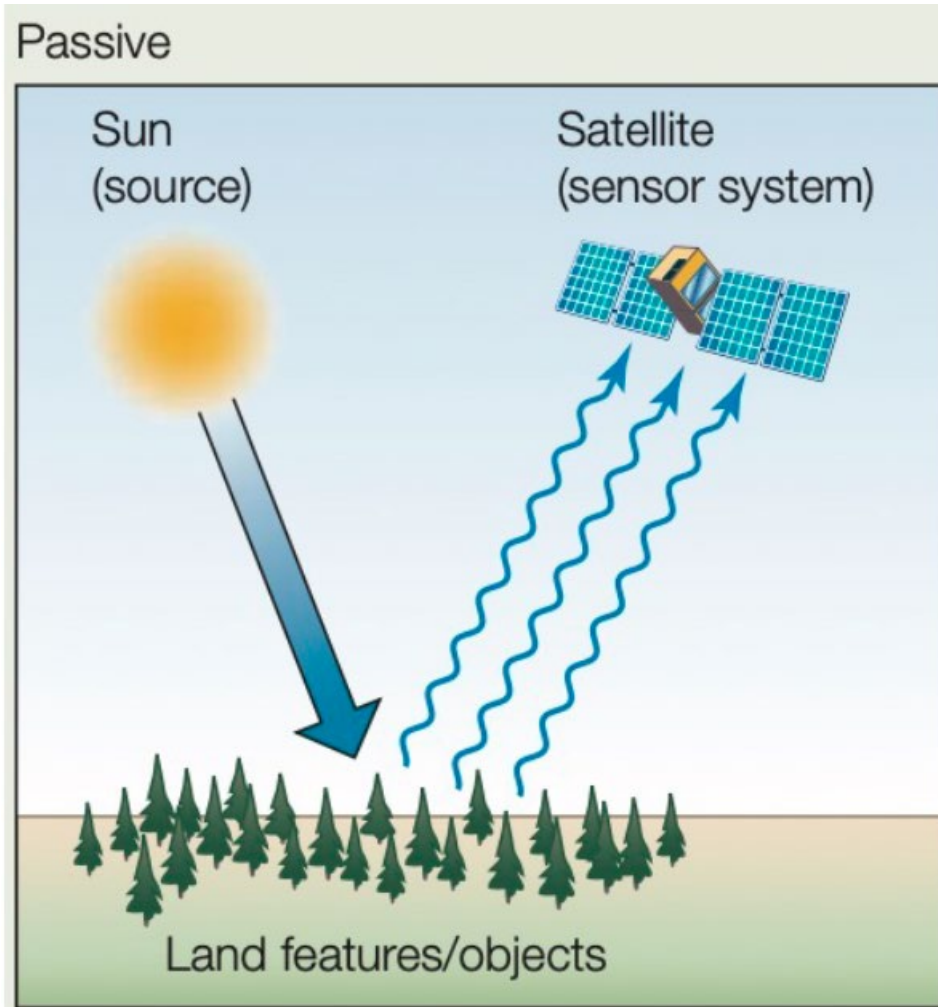




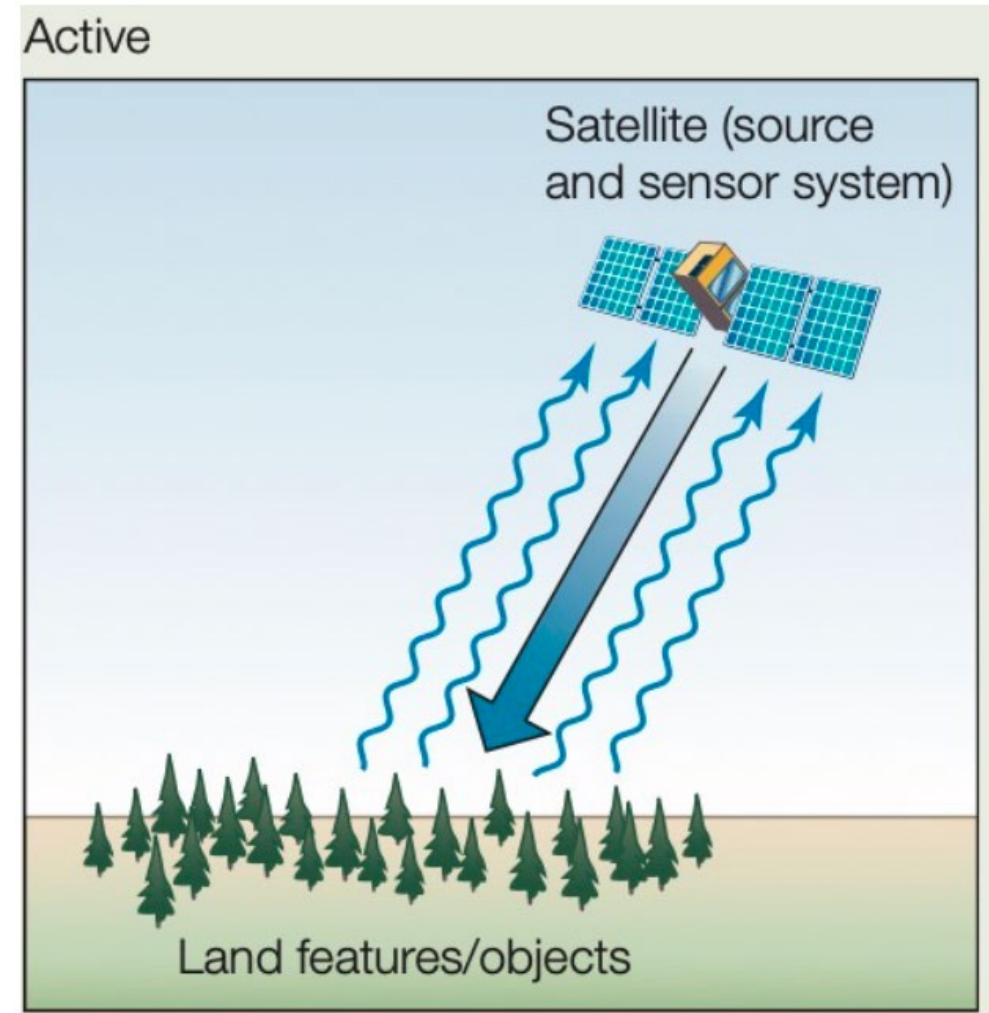


Key Concept

Active vs Passive Remote Sensing



Versus

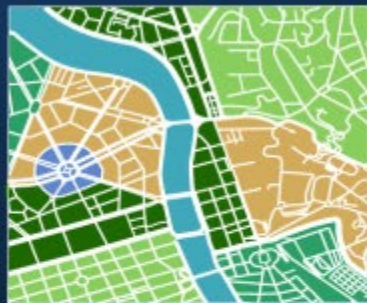


Key Concept

Spatial, Spectral, Temporal, and Radiometric Resolution

Remote Sensing Resolutions

SPATIAL RESOLUTION



1m pixel



30m pixel

SPECTRAL RESOLUTION

Wavelength ↑



Coarse - 4 bands



Fine - 8 bands

TEMPORAL RESOLUTION



Lower - Annual



Higher - Daily

RADIOMETRIC RESOLUTION

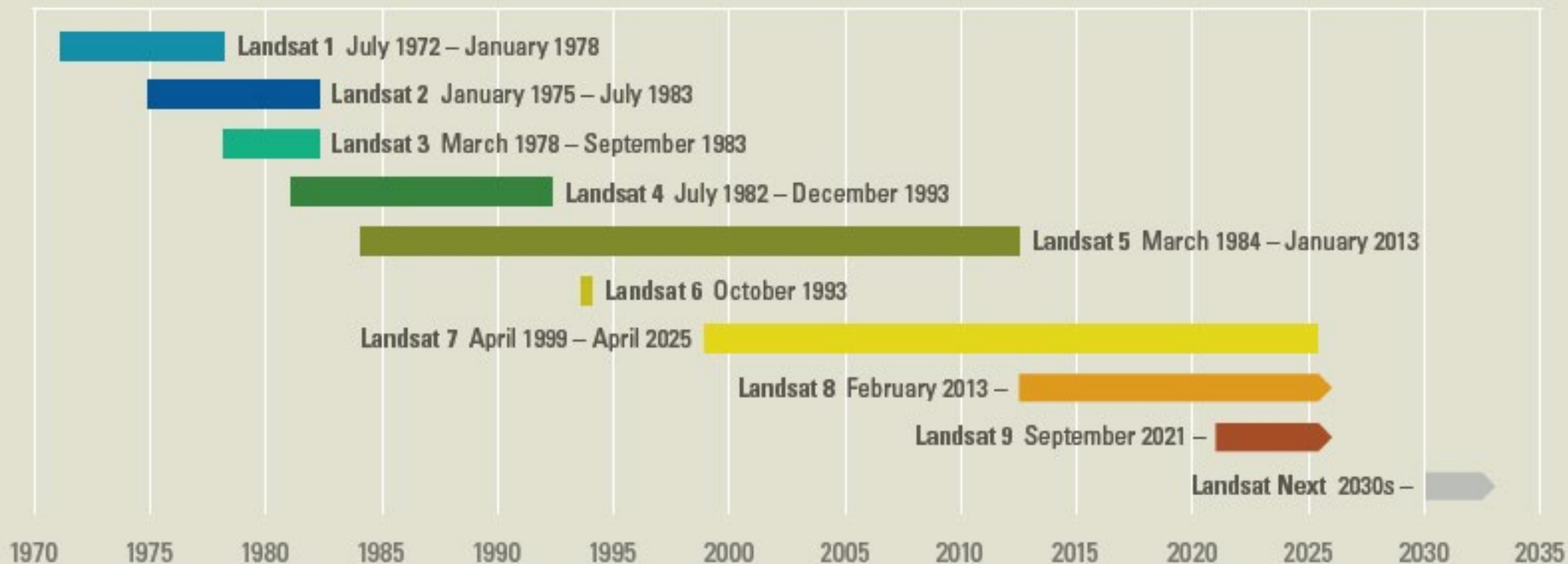


Coarse - 4 bit

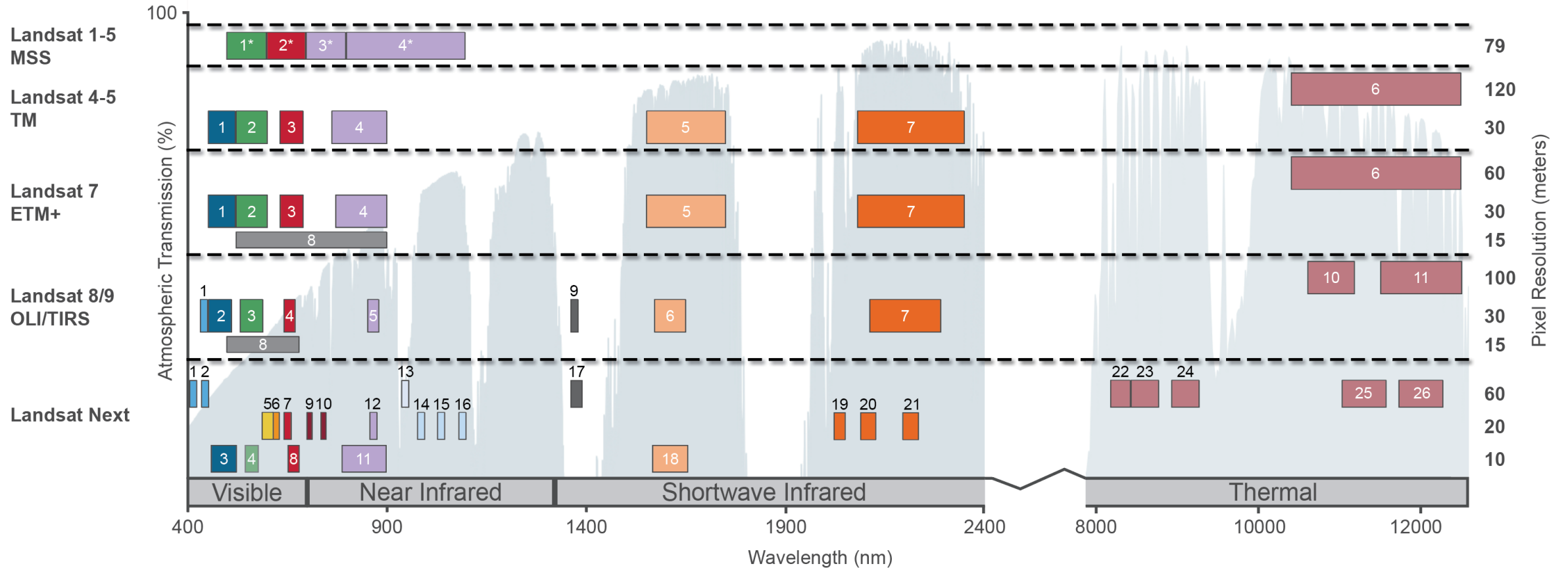


Fine - 8 bit

Landsat Missions: Imaging the Earth Since 1972



Comparison of Landsat Spectral Bands



* MSS bands 1–4 were known as bands 4–7, respectively, on Landsats 1–3

USGS, March 2024

NASA EARTH FLEET

OPERATING & FUTURE THROUGH 2023





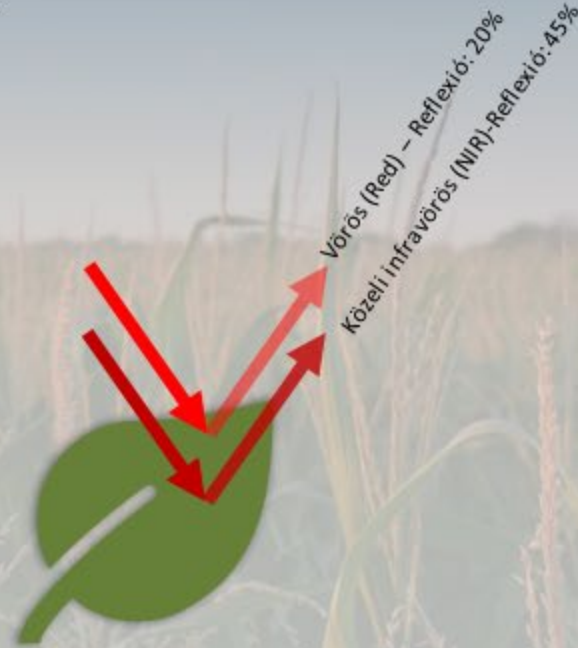
Calculation of NDVI

$$NDVI = \frac{NIR - RED}{NIR + RED}$$



Withered vegetation

$$NDVI = \frac{(0.40 - 0.35)}{(0.40 + 0.35)} = 0.07$$



Vegetation under stress

$$NDVI = \frac{(0.45 - 0.20)}{(0.45 + 0.20)} = 0.38$$



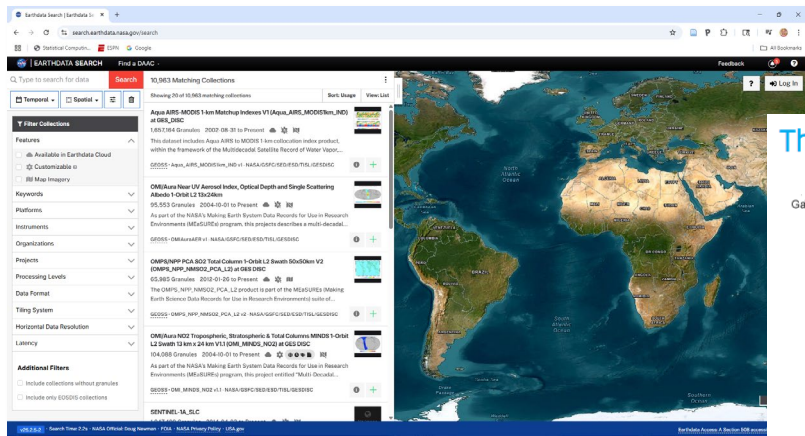
Healthy vegetation

$$NDVI = \frac{(0.60 - 0.09)}{(0.60 + 0.09)} = 0.74$$

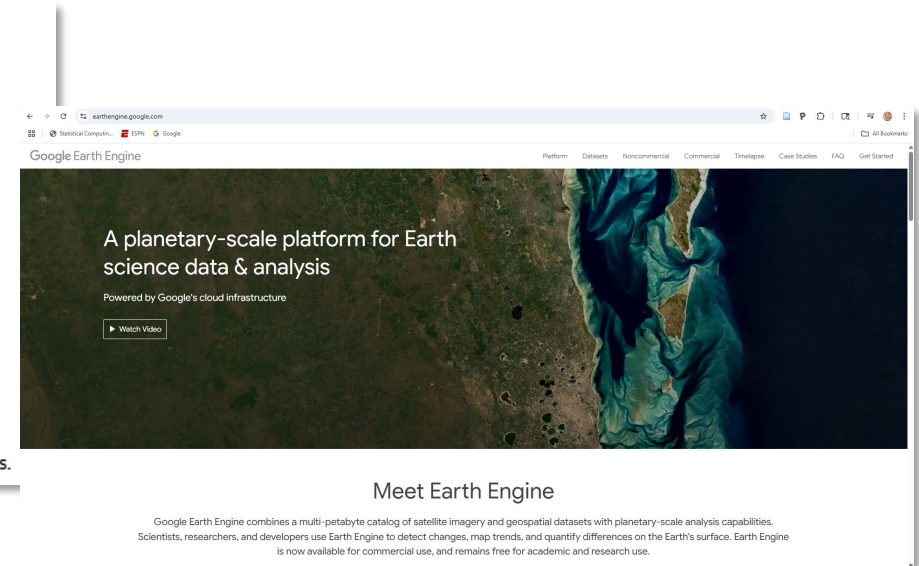
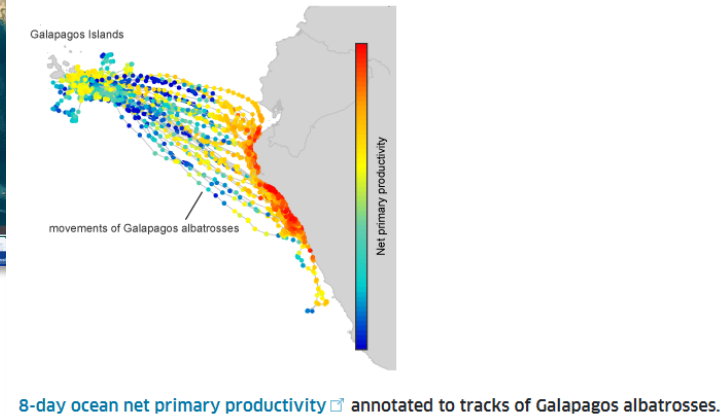
Where to Locate Satellite Data

Many sites exist to access data

- Navigate to:
 - NASA Earth Data - <https://search.earthdata.nasa.gov/search>
 - Movebank ENV Data Tool - <https://www.movebank.org/cms/movebank-content/env-data>
 - Google Earth Engine - <https://earthengine.google.com/>

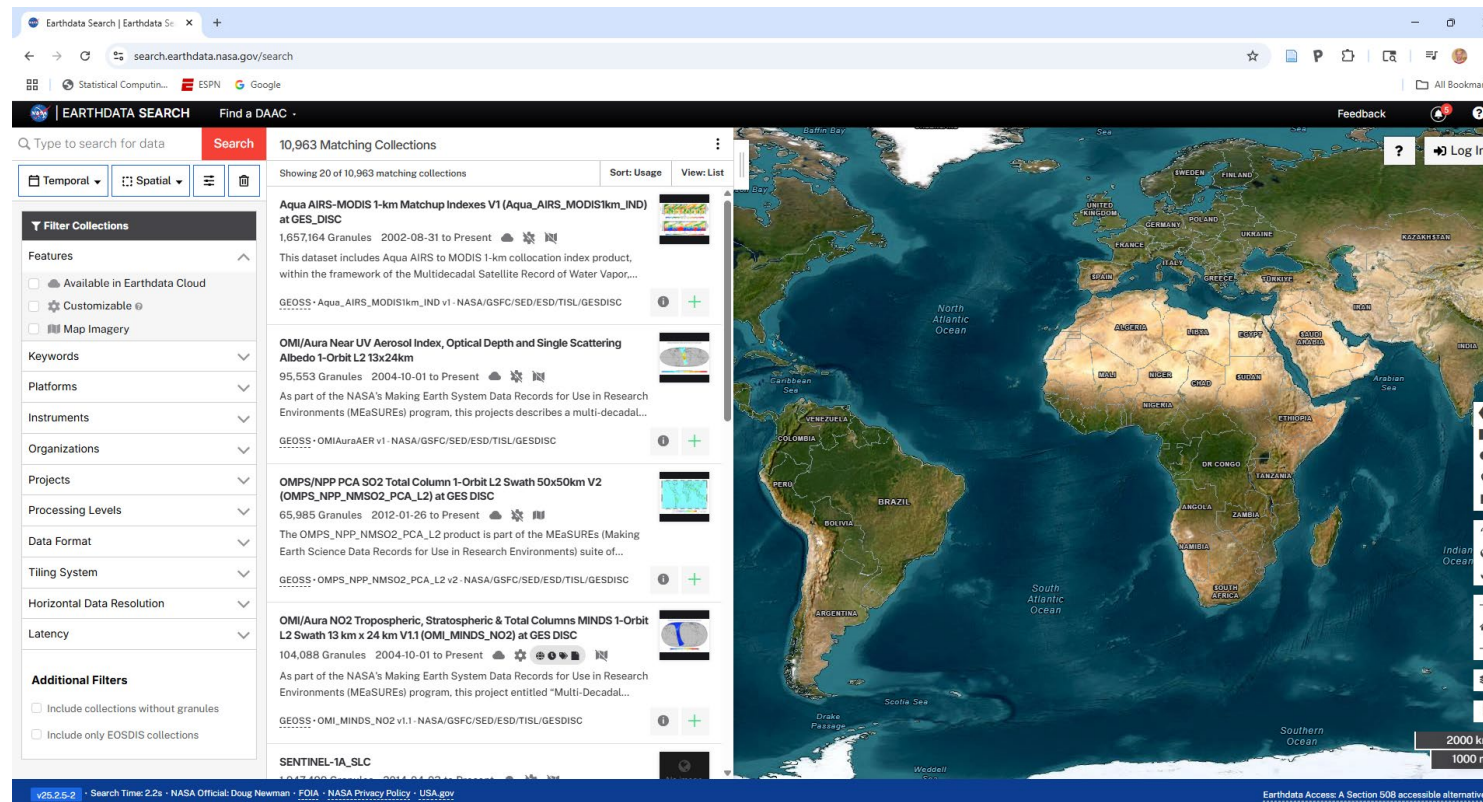


The Env-DATA System



NASA Earth Data

- Navigate to:
 - <https://search.earthdata.nasa.gov/search>



NASA Earth Data

- Enter Search Terms (vegetation, sea surface temperature, land cover, elevation, NDVI, ...)

The image displays the NASA Earth Data search interface. On the left, a sidebar shows the search bar with 'NDVI' entered, a 'Search' button, and filters for 'Temporal', 'Spatial', and 'Filter Collections'. The main area shows '237 Matching Collections' with a list of results including 'MODIS/Terra Vegetation Indices 16-Day L3 Global 250m SIN Grid V061', 'MODIS/Terra Vegetation Indices Monthly L3 Global 1km SIN Grid V061', and 'MODIS/Terra Vegetation Indices 16-Day L3 Global 500m SIN Grid V061'. A red box highlights the search bar and filters, with a red arrow pointing to the 'Filter Collections' section. On the right, a map of the world is shown with a scale bar indicating 2000 km and 1000 mi.

NASA Earth Data

- Select product

MODIS/Terra Vegetation Indices 16-Day L3 Global 250m SIN Grid V061

170,637 Granules 2000-02-18 to Present



The Terra Moderate Resolution Imaging Spectroradiometer (MODIS) Vegetation Indices (MOD13Q1) Version 6.1 data are generated every 16 days a...

GEOSS • MOD13Q1 v061 - LP DAAC



MODIS/Terra Vegetation Indices Monthly L3 Global 1km SIN Grid V061

88,939 Granules 2000-02-01 to Present



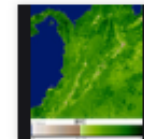
The Terra Moderate Resolution Imaging Spectroradiometer (MODIS) Vegetation Indices Monthly (MOD13A3) Version 6.1 data are provided monthl...

GEOSS • MOD13A3 v061 - LP DAAC



MODIS/Terra Vegetation Indices 16-Day L3 Global 500m SIN Grid V061

169,911 Granules 2000-02-18 to Present



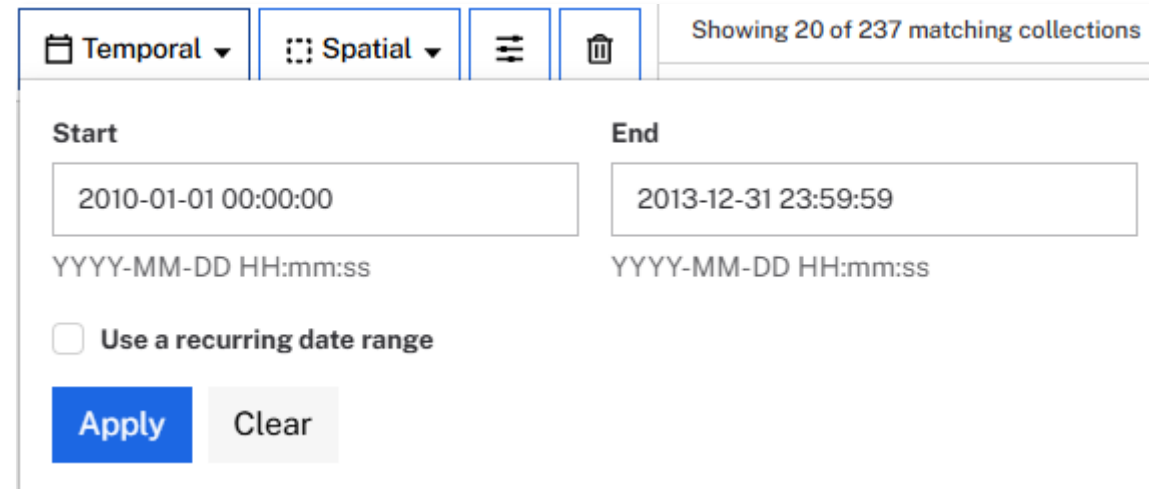
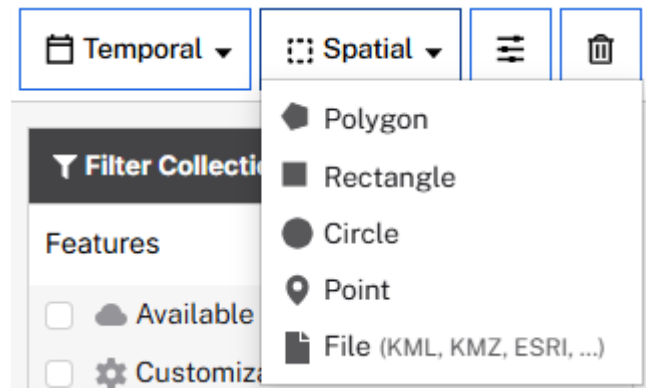
The MOD13A1 Version 6.1 product provides Vegetation Index (VI) values at a per pixel basis at 500 meter (m) spatial resolution. There are two primary...

GEOSS • MOD13A1 v061 - LP DAAC



NASA Earth Data

- Filter by spatial and temporal extent
Reduce the amount of data necessary to download



This screenshot shows the filter interface for NASA Earth Data. The 'Temporal' filter is selected, and the 'Spatial' filter is also visible. The 'Temporal' filter dropdown menu is open, showing options: Start, End, and Use a recurring date range. The 'Start' and 'End' date range is set from 2010-01-01 00:00:00 to 2013-12-31 23:59:59. The 'Apply' button is highlighted in blue, and the 'Clear' button is greyed out.

Showing 20 of 237 matching collections

Temporal **Spatial**

Start **End**

2010-01-01 00:00:00 2013-12-31 23:59:59

YYYY-MM-DD HH:mm:ss YYYY-MM-DD HH:mm:ss

☐ Use a recurring date range

Apply **Clear**

NASA Earth Data

- Check map and number of granules

The screenshot displays the NASA Earth Data search interface. The top navigation bar includes the NASA logo, 'EARTHDATA SEARCH', and a 'Find a DAAC' link. The search bar contains 'NDVI' and a red 'Search' button. Below the search bar, there are filters for 'Temporal' and 'Spatial'. The 'Spatial' filter is set to 'Rectangle' with coordinates SW: -2.18547, 36.62128 and NE: -1.25363, 37.36733. The 'Temporal' filter shows a 'Start' date of 2010-01-01 00:00:00 and a 'Stop' date of 2013-12-31 23:59:59. A 'Filter Granules' section is also visible, with a 'Clear Filters' button. The main search results area shows 'Search Results (82 Collections)' and 'MODIS/Terra Vegetation Indices 16-Day L3 Global 250m SIN Grid V061'. It indicates 'Showing 20 of 93 matching granules' and 'Sort: Start Date (Newest)'. The results are displayed in a grid of 10 items, each showing a thumbnail map, a 'START' date, an 'END' date, and a download icon. A 'Download All 93' button is at the bottom of the grid. On the right side, a map shows the search area in East Africa, with a blue rectangle highlighting the search region. The map includes labels for Nairobi, Langata, Machakos, Masai, Ollepesi, Kajiado, and Kajado. At the bottom, a timeline shows the years 1990, 2000, 2010, 2020, and 2030, with a green bar indicating the search period from 2010 to 2013.

NASA Earth Data

- Follow download instructions

The screenshot displays the NASA Earth Data Search interface. On the left, a sidebar shows the project details: "Untitled Project" with 93 Granules, 1 Collection, and 20.0 GB. The selected dataset is "MODIS/Terra Vegetation Indices 16-Day L3 Global 250m SIN Grid V061". The main panel shows the "Edit Options" section with two steps: "1 Choose how you want to download your data" and "2 Select a service and customize options". Under step 1, the "Download all data" option is selected, with a sub-option "Direct download of all selected data". Under step 2, a message states "No customization options are available for the selected access method." At the bottom left, there is a "Download Data" button. At the bottom right, it shows "Collection 1 of 1" and a "Done" button. On the right side of the interface, a satellite map of East Africa is displayed with a blue rectangular selection box covering a large area. The map includes labels for cities like Nairobi, Langata, Machakos, and Masii. At the bottom, a timeline slider is visible, ranging from 1990 to 2030, with a green bar indicating the data period from 1990 to approximately 2015.

NASA Earth Data

- Follow download instructions

Option 1: Use the Earthdata Download tool (<https://nasa.github.io/earthdata-download/>)

Option 2: Use command-line tools (“Download Script” tab)

Download Status

This page will automatically update as your orders are processed. The Download Status page can be accessed later by visiting <https://search.earthdata.nasa.gov/downloads/0246443449> or the [Download Status and History](#) page.

MODIS/Terra Vegetation Indices 16-Day L3 Global 250m SIN Grid V061

STATUS	ACCESS METHOD	GRANULES
Complete (100%) Updated: 07-27-2025 04:36:52 pm	Download	93 Granules

Download your data directly from the links below, or use the provided download script.

Download Files | AWS S3 Access | **Download Script** | Browse Imagery

Retrieved 363 files for 93 granules

100%

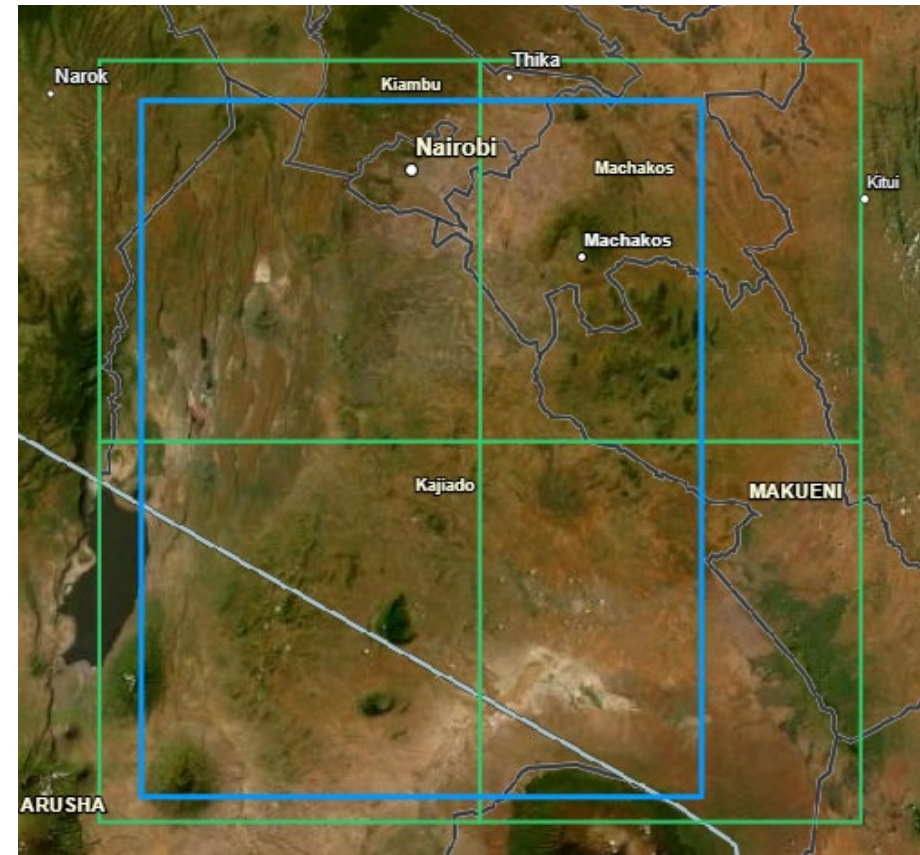
[Download Files](#) | [Copy](#) | [Save](#) | [Expand](#)

<https://data.lpdaac.earthdatacloud.nasa.gov/lp-prod-public/MOD13Q1.061/MOD13Q1.A2013353.h21v09.061.2021245>
<https://data.lpdaac.earthdatacloud.nasa.gov/lp-prod-protected/MOD13Q1.061/MOD13Q1.A2013353.h21v09.061.2021245>
<https://data.lpdaac.earthdatacloud.nasa.gov/lp-prod-public/MOD13Q1.061/MOD13Q1.A2013353.h21v09.061.2021245>

NASA Earth Data

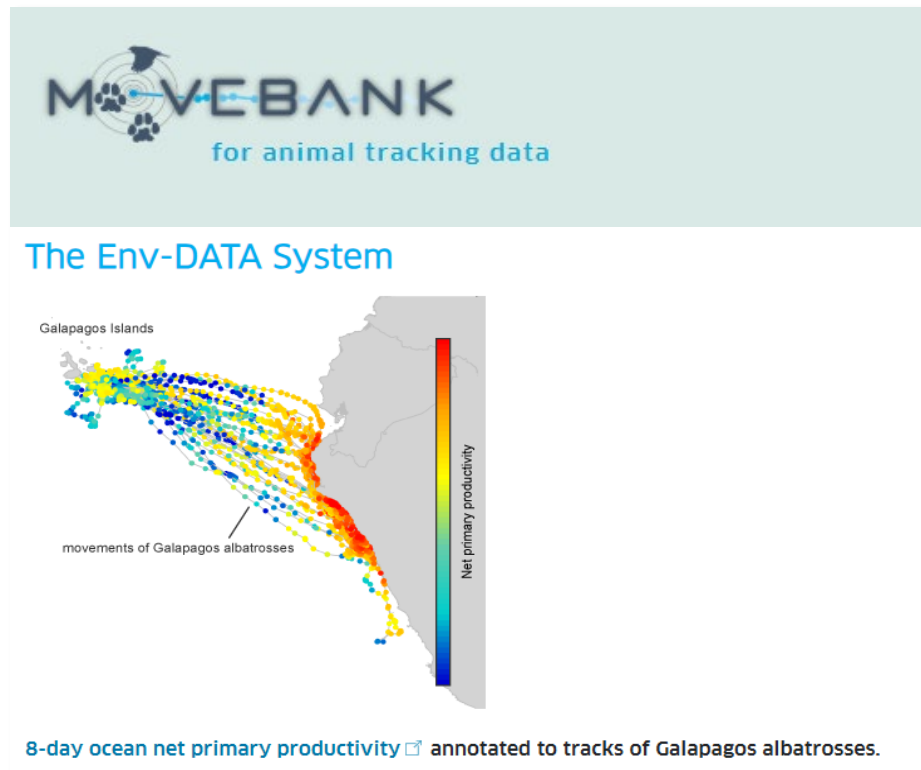
- Important Notes on Post-Processing
 - Many products will include multiple granules
 - E.g., Different locations or date/times
 - Many granules will include multiple layers
 - E.g., MODIS has multiple files for individual bands or projects (NDVI, EVI)
 - Requires GIS analyses
 - Mosaicking to combine spatial tiles
 - Stacking to combine temporal files
 - Cropping to include only need area

Example using SRTM elevation data: 4 tiles



Movebank ENV Data

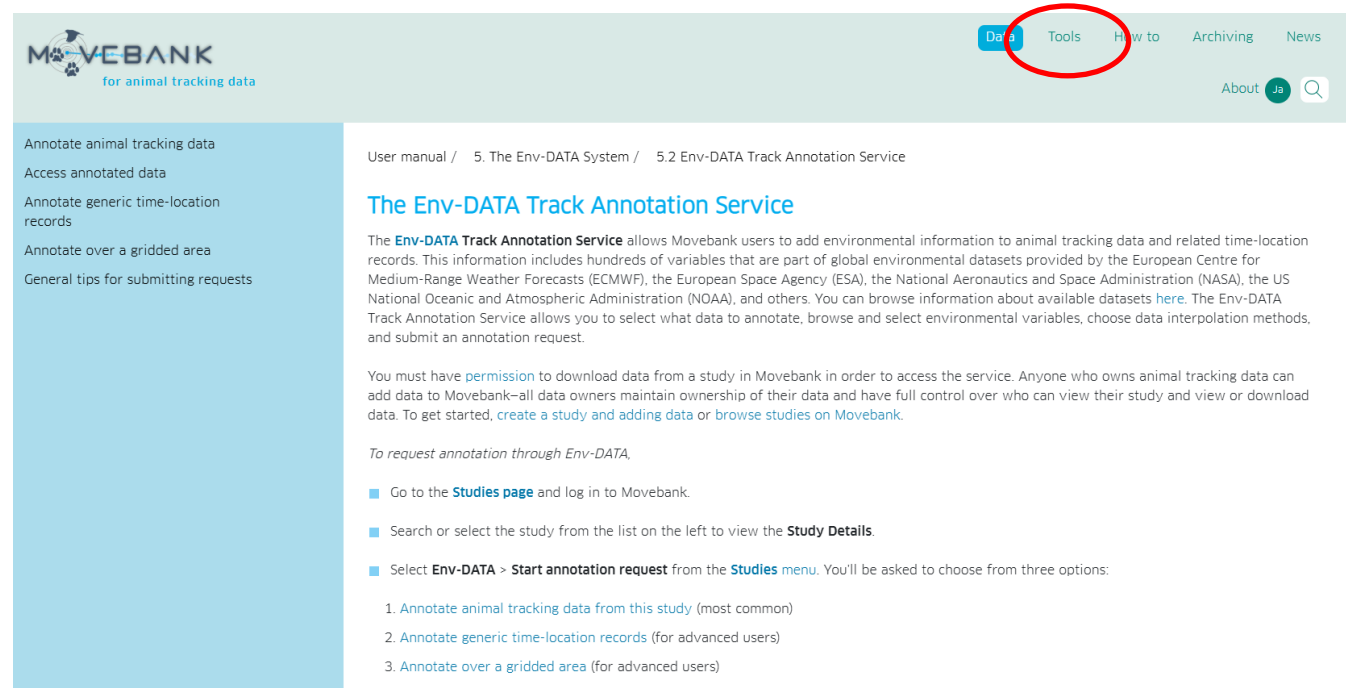
- Navigate to:
 - Movebank ENV Data Tool - <https://www.movebank.org/cms/movebank-content/env-data>



A set of free tools on Movebank to help you link animal movement data with information from global environmental datasets

Movebank ENV Data

- Go to Tools – EnvDATA:
 - Choose from:
 - Annotate animal tracking data from this study (most common) ←
 - Annotate generic time-location records (for advanced users)
 - Annotate over a gridded area (for advanced users)



Movebank ENV Data

- Start Annotation Request:
 - Go to **Study** and select **Start Annotation Request**

The screenshot displays the Movebank website for animal tracking data. The header features the Movebank logo and the tagline 'for animal tracking data'. Below the header, there is a navigation bar with tabs for 'Search', 'Studies', and 'Env-DATA'. The 'Studies' tab is active, showing a list of studies on the left and details for the selected study on the right. The selected study is 'White-bearded wildebeest (Connochaetes taurinus) movements - Kenya'. The 'Start Annotation Request' button is circled in red in the top right corner of the 'Studies' tab.

Movebank
for animal tracking data

Search Studies

Studies where I am Data Manager

Filter by Study Name

Study - White-bearded wildebeest (Connochaetes taurinus) movements - Kenya
[Animals](#) [Tags](#) [Files](#) [Argos Feeds](#) [File Formats](#)

[Addax \(Addax nasomaculatus\) reintroduction - Morocco](#)
[African elephant \(Loxodonta africana\) - Namibia \(Kilian\)](#)
[Bobcat \(Lynx rufus\) - Virginia](#)
[Coyote \(Canis latrans\) movement in the northern blue ridge mountains -](#)
[Jaguar \(Panthera onca\) Movements - Brazil](#)
[Korrigum \(Damaliscus lunatus korrigum\) - Benin](#)
[Maned wolf \(Chrysocyon brachyurus\) movements - Brazil](#)
[Mountain Tapir \(Tapirus pinchaque\) movements - Ecuador](#)
[Plains Bison \(Bison bison\) - Montana \(Shamon\)](#)
[Przewalski Horse \(Equus przewalskii\) - Mongolia \(Songer\)](#)
[Scimitar-horned oryx \(Oryx dammah\) reintroduction - Chad](#)
[Western Hartebeest \(Alcelaphus buselaphus major\) - Benin](#)
[White-bearded wildebeest \(Connochaetes taurinus\) - Greater Mara Eco](#)
[White-bearded wildebeest \(Connochaetes taurinus\) - Greater Mara Eco](#)
[White-bearded wildebeest \(Connochaetes taurinus\) movements - K](#)

Study Details

Study Name White-bearded wildebeest (Connochaetes taurinus) movements - Kenya

Contact Person Jared

Principal Investigator Randall Boone

PI Contact Details 1499 Campus Delivery Natural Resource Ecology Laboratory
Colorado State University Fort Collins, CO 80523-1499
Stabach JA, Hughey L, Reid RS, Worden JS, Leimgruber P, Boone RB. 2020. Data from: Study "White-bearded wildebeest in Kenya". Movebank Data Repository. <https://www.doi.org/10.5441/001/1.h0t27719>

Citation Walton T, Boone R. 2017. Analyzing wildebeest behavior and effects of humans on activity in southwest Kenya. Poster [stored in this study; see Files]
Stabach JA. 2015. Movement, resource selection, and the physiological stress response of white-bearded wildebeest [dissertation]. [Fort Collins (CO, USA)]: Colorado State University. <https://doi.org/10.217/167207>
Boone RB, Reid RS, Lilieholm RJ, Worden JS, Ogutu JO. 2009. Wildebeest forage acquisition in fragmented landscapes under variable climates. National Science Foundation (NSF) DEB Grant 0810292

[Start Annotation Request](#)
[Show My Requests](#)
[About Env-DATA](#)

Movebank ENV Data

- Annotate animal tracking data

- Choose what animal tracking data you want to annotate. You can select from any of the Tags or Animals in the study, and can choose to include outliers and limit the annotation to one sensor type. When you are done select **Continue**.

Choose which data from this study you want to annotate

☒ Select animals

☐ Select tags

Available sensor types

GPS ▾

☐ Include outliers

Filter by date

Interval:

select interval ▾

From:

To:

☒ Nenkongu Narok

☒ Nentepesi

☒ Nesimej

☒ Nkairowua

☒ Nkoko

☒ Noontare

☒ Ntishya

☒ Ole Nagol

☒ Olope

☒ Paita

☒ Peria

☒ Reiyo

☒ Sawani

☒ Sotua

☒ Taletian

Select all

Deselect all

Cancel

Continue

Movebank ENV Data

- Annotate animal tracking data

- Next, browse and select the environmental variables you want to annotate. You can browse by [source](#)—such as MODIS Ocean or the ECMWF weather model—or by variable type—such as elevation or weather. To learn more about individual data sources or variables, click on ⓘ to view more information and links. **If you are requesting variables from multiple sources, we recommend splitting these into multiple smaller requests.**

Browse available environmental variables and select those you want to annotate

Browse available variables based on their source provider or the type of information they provide. If you will annotate many variables, we recommend submitting separate requests for each. The service is indicated in the first part of the name of selected variables. [User manual](#)

Variables by sourceVariables by type

⊕ ASTER ASTGTM3 Global 30-m DEM ⓘ

⊕ ETOPO1 1-arc-minute Global Relief Model ⓘ

⊕ GlobCover

⊕ MODIS Land

⊕ MODIS Ocean

⊕ Movebank Derived Variables

⊕ NASA Distance to the Nearest Coast ⓘ

⊕ NASA Socioeconomic Data and Applications Center (SEDAC)

⊕ NCEP North American Regional Reanalysis (NARR)

⊕ NOAA Global Climate Indexes ⓘ

⊕ Oregon State Ocean Productivity Reanalysis

⊖ SRTM 1 Arc-Second DEM ⓘ

Variable

SRTM Elevation

Remove

Cancel

Back

Continue

Derived from information provided by NASA's ARSET program – Introduction to the Integration of Animal Tracking and Remote Sensing

Movebank ENV Data

- Review and Submit (Send Annotation Request)
 - Data requests are available for download for **two** weeks as a .zip file
 - Annotated data will be shown as **available** in the **My requests** table (under **Env-Data** > **Show my requests** in the **Studies** menu).

Review a summary of your order and submit your annotation request

study:

White-bearded wildebeest
(Connochaetes taurinus)
movements - Kenya

sensor type:

GPS

tracking data:

Animal: Fifteen, Karbolo, Kayioni, Kikaya, Kiranto, Koyiaki, Laingoni, Ledama, Lempaash, Lengaruni, Leporana, Lerisa, Motorogi, Naboisho, Nagol, Naiba, Naisula, Naserian, Ne Mpaash, Neatoo, Nenkitirua, Nenkongu Narok, Nentepesi, Nesime, Nkairowua, Nkoko, Noontare, Ntishya, Ole Nagol, Olope, Paita, Peria, Reiyo, Sawani, Sotua, Taletian

Send download notification to:

jared_stabach@hotmail.com

Request name:

White-bearded wildebeest (Connochaetes t

Variable	Interpolation
SRTM Elevation	bilinear

Cancel

Back

Send annotation request

Note: Movebank requests can be time consuming. Best to split your data into smaller parts (i.e., create data subsets (a few animals) and request fewer environmental datasets (one or two) for annotation)

Movebank ENV Data

- Review and Submit (Send Annotation Request)
 - Data requests are available for download for **two** weeks as a .zip file
 - Annotated data will be shown as **available** in the **My requests** table (under **Env-Data** > **Show my requests** in the **Studies** menu).

MOVEBANK
for animal tracking data

Search Studies

Studies where I am Data Manager New Study

Filter by Study Name

Study - White-bearded wildebeest (Connochaetes taurinus) movements - Kenya

Animals Tags Files Arcs Feeds File Formats

Study Details

Study Name White-bearded wildebeest (Connochaetes taurinus) movements - Kenya

Contact Person Jared Randall Boone

Principal Investigator

PI Contact Details

1499 Campus Delivery Natural Resource Ecology Laboratory
Colorado State University Fort Collins, CO 80523-1499

Stabach JA, Hughey L, Reid RS, Worden JS, Leimgruber P, Boone RB. 2020. Data from: Study "White-bearded wildebeest in Kenya". Movebank Data Repository
<https://www.doi.org/10.5441/001/1.h0127719>

Citation

Walton T, Boone R. 2017. Analyzing wildebeest behavior and effects of humans on activity in southwest Kenya. Poster [stored in this study; see Files]

Stabach JA. 2015. Movement, resource selection, and the physiological stress response of white-bearded wildebeest [dissertation]. [Fort Collins (CO, USA)]: Colorado State University. <https://doi.org/10.1021/167207>

Boone RB, Reid RS, Lilieholm RJ, Worden JS, Ogutu JO. 2009. Wildebeest forage acquisition in fragmented landscapes under variable climates. National Science Foundation (NSF) DEB

My requests

[User manual](#)

Find a list of your submitted annotation requests below. Click on the links in the table for information on submitted requests. Once results are available, download annotated data from the link on the right side of the table. Processing time depends on the spatial and temporal scope and resolution of your dataset and the requested environmental attributes, the number of other requests being processed and whether others have recently requested the same environmental data. If your request fails or you have not gotten your results within a few weeks, please send the access key to support@movebank.org for assistance. You can copy the access key from under **Details** (third paragraph) below. [Read more](#) for guidance on interpreting your results and answers to frequently asked questions.

Request	Timestamp	State	Overview	Details
White-bearded wildebeest (Connochaetes taurinus) movements - Kenya	2025-07-28 12:13:52.670	available	view	view download

[Close](#) [Refresh](#)

Note: Movebank requests can be time consuming. Best to split your data into smaller parts (i.e., create data subsets (a few animals) and request fewer environmental datasets (one or two) for annotation)

Movebank ENV Data – Other Applications

- Annotate generic time-location records

Get environmental information for times and locations related to but not the same as the animal location themselves (e.g., simulated trajectories or randomly sampled locations)

Upload data for annotation

Here you can request annotation of generic time-location records related to your tracking data on Movebank. Select from a list of previously submitted files or upload a new file.

Uploaded files must be comma separated with a header including the following fields:

- timestamp (required, date and time in UTC in this format: yyyy-MM-dd HH:mm:ss.SSS)
- location-lat (required, longitude projected in WGS84 in decimal degrees, range -180 to 180)
- location-long (required, latitude projected in WGS84 in decimal degrees)
- height-above-msl (optional, height above mean sea level in meters)

Other fields are allowed but will not be evaluated for annotation. As noted above, all calculations will assume that times are in UTC, location coordinates are projected in WGS84, and heights represent height above mean sea level. Height above ellipsoid needs to be converted before submission. Null values should be left blank. For additional information see [here](#)

[Download example data](#)

Upload file

Choose File

No file chosen

Cancel

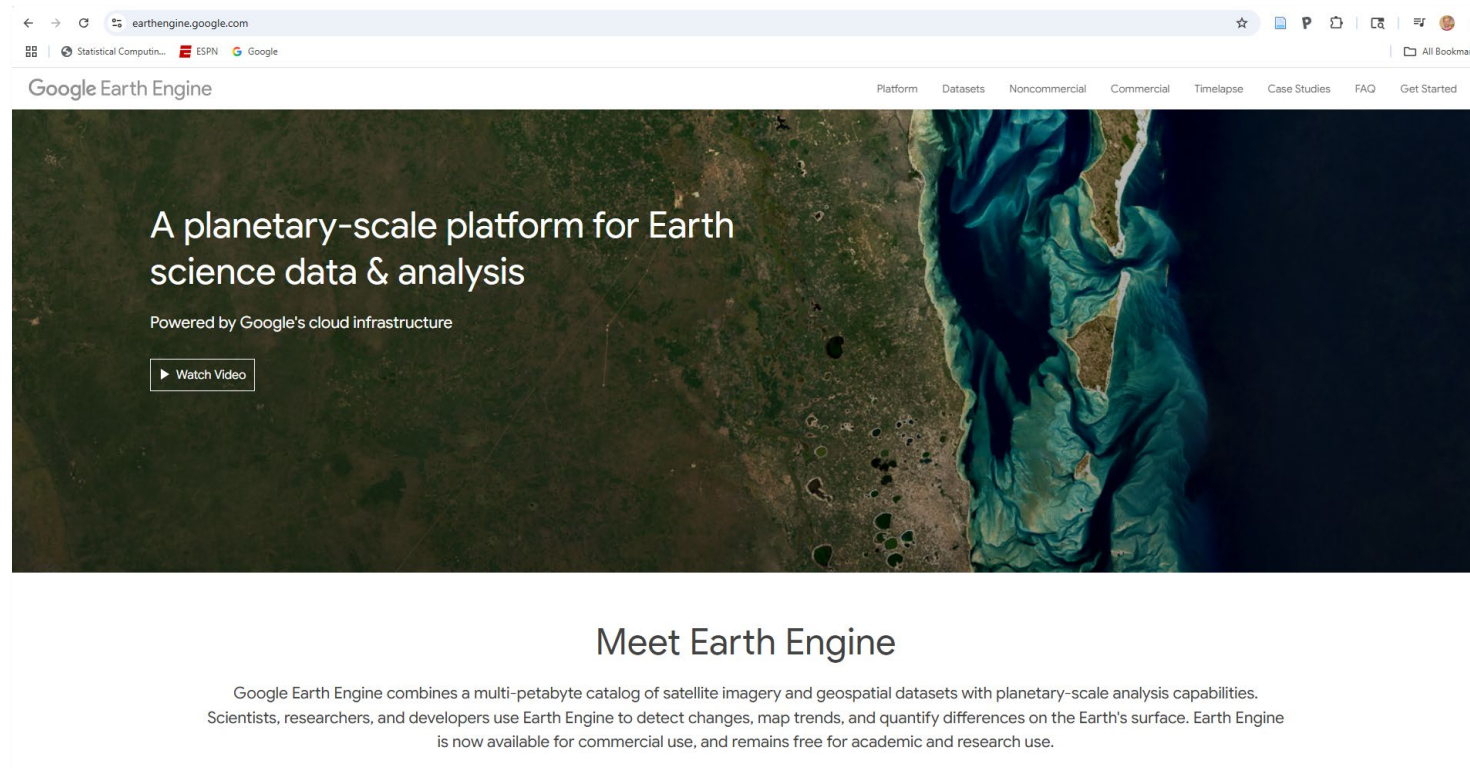
Continue

Note: Make sure the uploaded files matches the required format **EXACTLY**

Google Earth Engine

A Multi-petabyte catalog and geospatial datasets with planetary-scale analysis capabilities

- Navigate to:
 - Google Earth Engine - <https://earthengine.google.com/>



Google Earth Engine

- Follow GEE workflow

Open Access Communication

Enhancing Animal Movement Analyses: Spatiotemporal Matching of Animal Positions with Remotely Sensed Data Using Google Earth Engine and R

by Ramiro D. Crego ^{1,2,*}, Majaliwa M. Masolele ³, Grant Connette ^{1,2} and Jared A. Stabach ¹

